

A
Major Project Report
On
**Sparse Machine Learning Approach for Biomarker Detection in
Alzheimer's Disease**

Submitted in partial fulfillment of the
Requirements for the award of the degree

IN
MASTER OF COMPUTER APPLICATIONS

Submitted By
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DEPARTMENT OF INFORMATION TECHNOLOGY



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(2024 – 2026)

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CERTIFICATE

This is to certify that the project report entitled “**Sparse Machine Learning Approach for Biomarker Detection in Alzheimer’s Disease**” is a Bonafide work done and submitted by **GAJAVALLI VINAY KUMAR GUPTA (24MC201A97)** in partial fulfilment of the requirements for the award of the degree of MASTER OF COMPUTER APPLICATIONS from **Anurag University** during the academic year 2025-2026. The Bonafide work has not been submitted elsewhere for the award of any other degree.

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ACKNOWLEDGEMENT

I would like to express my profound sense of gratitude to all for having helped me in completing this dissertation. I would like to express my deep-felt gratitude and sincere thanks to my guide, **Dr. Niteesha Sharma**, Head of the Department (IT), of Anurag University, for **her** skilful guidance, timely suggestions, and encouragement in completing this project.

A special acknowledgment is extended to **Dr. Niteesha Sharma**, Head of the Department (IT), for her valuable support, and to **Mr. G. Sekhar Reddy**, Program Coordinator (MCA), for his guidance, which significantly contributed to the successful completion of this project.

I would like to express my sincere thanks to **Dr. K. Sudheer Reddy**, Dean of Academics and Planning, Anurag University, whose motivation in the field of software development has made me overcome all hardships during the study and successful completion of the project.

I extend my sincere appreciation to **Dr. V. Vijaya Kumar**, Dean, School of Engineering, for his encouragement and consistent support during this endeavor.

Finally, I would like to express my heartfelt thanks to my parents, who were very supportive both financially and mentally and for their encouragement to achieve my set goals.

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DECLARATION

This is to certify that the major project titled **“SPARSE MACHINE LEARNING APPROACH FOR BIOMARKER DETECTION IN ALZHEIMER’S DISEASE”** is a Bonafide work done by me in fulfillment of the requirements for the award of the degree **MASTER OF COMPUTER APPLICATIONS** and submitted to **Anurag University, Hyderabad**. I also declare that this project is a result of my own effort and has not been copied or intimated from any source. Citations from any websites are mentioned in the bibliography. This work was not submitted earlier at any other university for the award of any degree.

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ABSTRACT

Alzheimer’s Disease (AD) is a neurological condition associated with progressive impairment of cognitive and memory abilities in older adults. Detecting AD at early stages is critical for appropriate clinical treatment and diagnosis. This work presents a new Temporal Group Sparse Additive Learning (TGSAL) approach to detect Alzheimer’s Disease through neuroimaging data with both longitudinal and cross-sectional forms. The proposed algorithm involves a feature elimination process that involves eliminating highly missing features, conducting median imputation, and scaling using standardization. In the first stage, the algorithm uses an adaptive LASSO method to select discriminative features among different classes. In the second stage, a Group LASSO approach considers the anatomy of the brain by considering entire brain regions such as hippocampi, entorhinal cortex, amygdala, temporal lobes, and frontal lobes. For the modeling of nonlinear relations between features, Sparse Additive Model combined with B-spline basis expansion and Ridge regression is employed, thus allowing for flexibility of the model with the overfitting prevention provided by sparsity constraints. Moreover, a Temporal Sparse Regression component is implemented for obtaining the progression biomarkers from longitudinal data, calculating rate-of-change values which are stable over time points. The resulting feature subset with nonlinear and progression features is fed into the XGBoost classifier tuned with appropriate parameters. As shown by experimental results, the suggested method succeeds in the dimensionality reduction of feature space along with the preservation of the high prediction quality measured in terms of accuracy and ROC-AUC values. The method can be used to interpret biomarkers relevant to certain brain regions affected by Alzheimer’s Disease.

Keywords: Alzheimer’s Disease, Sparse Learning, Neuroimaging Biomarkers, MRI, ADNI Dataset, Feature Selection, XGBoost, Temporal Learning, k fold Cross Validation

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LIST OF ABBREVIATIONS

S.No.	Full Form	Abbreviation
1	Alzheimer's Disease	AD
2	Cognitively Normal	CN
3	Mild Cognitive Impairment	MCI
4	Alzheimer's Disease Neuroimaging Initiative	ADNI
5	Open Access Series of Imaging Studies	OASIS
6	Magnetic Resonance Imaging	MRI
7	Machine Learning	ML
8	Extreme Gradient Boosting	XGBoost
9	Least Absolute Shrinkage and Selection Operator	LASSO
10	Group Least Absolute Shrinkage and Selection Operator	Group LASSO
11	Temporal Group Sparse Additive Learning	TGSAL
12	Receiver Operating Characteristic	ROC
13	Area Under the Curve	AUC
14	Precision-Recall Trade-off Measure	F1-Score
15	Z-score Normalization	Z-Norm

1. INTRODUCTION

1.1 Overview of The Domain

Alzheimer's disease (AD) is an incurable neurodegenerative disease that deteriorates cognition, memory, and behavior of the patient. It is considered to be one of the most common reasons of dementia among the elderly people of the world. The increase in longevity results in a significant growth of Alzheimer's disease prevalence rate.

Conventional techniques use clinical evaluation, cognitive assessment (MMSE and ADAS-Cog), and neuroimaging (Magnetic Resonance Imaging) for diagnosing Alzheimer's disease. These methods are time-consuming and require subjectivity in their application. In addition to these issues, neural imaging data is characterized by high dimensionality and complexity.

In order to resolve the described drawbacks, researchers widely apply machine learning methods for detecting Alzheimer's disease based on neuroimaging and clinical data. Support Vector Machine, Random Forest and some deep learning approaches were successfully used to classify subjects suffering from this disease. Unfortunately, machine learning-based approaches suffer from several drawbacks, which include high-dimensionality, redundancy, lack of interpretability, and weak generalizability.

Modern approaches highlight the importance of employing methods related to sparse learning and temporal modeling to handle these problems. The application of methods such as sparse learning allows selecting appropriate features by removing noise from the dataset. Temporal modeling allows understanding the process of the development of the disease. It is worth mentioning that the Temporal Group Sparse Additive Learning (TGSAL) approach provides the combination of feature selection, anatomical grouping, nonlinear modeling, and temporal modeling.

This project will be based on the idea and employ a structured sparse learning pipeline to classify patients with Alzheimer's Disease.

1.2 Objective of the project

The first and foremost aim of this study is to develop a predictive machine learning classifier that utilizes the neuroimaging data for the diagnosis of Alzheimer's Disease. The classification task will be a two-class problem, where the samples are classified as either AD or not. This implies the existence of one positive class, i.e., AD, and one negative class,

namely, Rest, which consists of both CN and MCI groups. The current research concentrates on processing high-dimensional neuroimaging data, designing an appropriate feature matrix, and applying the multi-stage sparse learning strategy.

In order to achieve the above-mentioned goal, a series of steps is suggested: Adaptive LASSO for feature selection, anatomical group-based sparsity for the preservation of relevant features, sparse additive learning for the consideration of possible nonlinear dependencies between features and classes, and temporal sparse regression as a tool to model the evolution of a disease over time. Finally, the developed sparse models will be used as a basis for a predictive classifier implemented based on the state-of-the-art XGBoost algorithm. The advantage of using XGBoost lies in the ability to handle complex nonlinear associations between input variables and output labels and also manage class imbalances well.

1.3 Problem Formulation

The Alzheimer's Disease classification is stated to be a supervised binary classification problem on high dimensional neuroimaging data. The data set contains features that characterize changes in various brain regions obtained from an MRI scan of the patients alongside their diagnosis label. In other words, the goal is to map the patients into two categories, either having Alzheimer's Disease (AD) or not (Rest). One of the main difficulties in the described classification problem arises due to the fact that the number of dimensions is extremely high while the number of samples is relatively low. It results in problems like overfitting and poor generalization of the learned models. Incomplete data and skewed distribution in medical databases are some other factors that make the problem of classification difficult to solve. The next challenge is related to the non-linear connection between features and diseases that can only be modelled using advanced techniques. Moreover, it should be noted that Alzheimer's disease has a temporal aspect, and not taking this into account will make the analysis incomplete. In order to overcome the aforementioned challenges, it is planned to use the multi-stage framework of sparse learning for classification purposes.

1.4 Scope

This research project addresses the task of constructing a framework for the classification of Alzheimer's Disease with the use of structured neuroimaging data sets. Mainly, the problem addressed here revolves around the binary classification of Alzheimer's Disease, i.e.,

identifying whether the patient belongs to the AD class or the non-AD class. Some of the data sets used in this research project are the ADNI data set, OASIS data set, and the Kaggle data set, providing information from the MRIs as well as the patients' clinical details.

In this study, the system uses a hierarchical approach in learning to identify important neuroimaging markers associated with Alzheimer's Disease, such as hippocampus, temporal lobes, and the entorhinal cortex. This approach allows the use of temporal modeling, whereby temporal information on the disease can be extracted and studied. However, the project does not deal with real-time application of this system nor does it use raw images in deep learning algorithms.

1.5 System Requirements

Implementation of the suggested Alzheimer's disease classification method depends on having a clearly defined set of software and hardware requirements. As the suggested classification approach relies on multiple steps of the sparse machine learning algorithm using high-dimensional data obtained from neuroimages, the system should provide efficient performance for the whole pipeline, from the initial data pre-processing to the classification step itself. These requirements are formulated to enable effective data processing of large amounts of data (such as ADNI and OASIS data), efficient use of sparse machine learning algorithms and training the XGBoost classifier. The developed solution does not need any special hardware or software environment.

1.5.1 Software Requirements

- Programming Language: Python
- Environment: Jupyter Notebook/Google Colab
- Packages:
 - NumPy – Numerical calculations
 - Pandas – Data processing
 - Scikit-Learn – Data processing and Model Evaluation
 - XGBoost – Classifier Model
 - Matplotlib/Seaborn – Graphing
- Operating System: Windows/Linux

- Data Processing: Capable of handling neuroimaging data from high dimensional space (ADNI, OASIS dataset)
- Model Category: Machine Learning
- Feature Engineering: Capable of handling sparse learning and feature selection

1.5.2 Hardware Requirements

- Processor: **Intel i5 or higher**
- RAM: **Minimum 8 GB (16 GB recommended)**
- Storage: **At least 10 GB free space**
- GPU: **Optional (not mandatory)**
- System Type: **Standard laptop/desktop sufficient**
- Processing Capability: Supports **multi-stage sparse learning pipeline**
- Suitable for handling **medium to large datasets**

2. LITERATURE SURVEY

2.1 Overview of Literature Survey

A lot of studies have been performed on the topic of Alzheimer's Disease (AD) utilizing various computational methods. Machine Learning (ML) and neuroimaging studies are currently among the major fields that are being considered due to the availability of large scale databases like Alzheimer's Disease Neuroimaging Initiative (ADNI). Various types of models and algorithms are analyzed with the goal of better predicting early diagnostics and disease progression. In this literature survey, we consider the problem of AD classification by analyzing approaches suggested for addressing it with the use of ML-based modeling.

The vast majority of works considers classifying the patient based on certain cognition stage and making predictions about disease progression using various types of imaging (structural MRI and PET). As neuroimaging is complex and high-dimensional, designing models requires addressing certain difficulties such as incorporating temporal information and selecting features. Recent research directions focus on developing more robust models using feature selection, dimensionality reduction, and incorporation of temporal information in multi-stage frameworks.

This literature survey covers a variety of works describing various ways of solving the problem of AD classification.

2.2 Importance of Literature Survey

The literature review is highly crucial in providing information regarding the current status as well as shortcomings associated with Alzheimer's Disease. It provides a lot of information regarding the development of computational approaches utilized in the identification process. This allows for the identification of areas of interest, which require more research. Additionally, literature reviews allow one to assess the efficiency of particular methods as well as determine whether the methods are suitable for practical use.

When it comes to the issue of neuroimaging datasets utilized in Alzheimer's Disease, a literature review proves to be very crucial. The problem involves data, which is usually high dimensional, and consists of irrelevant variables. In order to overcome such a challenge, a number of studies have employed feature selection techniques and sparse learning, among others. In addition, the research also emphasizes the importance of using time-related data in

predicting the progression of disease. Since Alzheimer's Disease is a progressive one, it is very important to consider any changes that take place during its progress. But most of the existing models lack the ability to capture the temporal data. It is also found from the literature review that even though deep learning models produce highly accurate predictions, these models need a lot of data for training and do not have good interpretability.

From all of the above, it can be said that the literature review helps us find out the advantages and disadvantages of existing models.

2.3 Literature Review

The aim of the literature review below is to evaluate different research studies related to AD diagnosis by means of applying machine learning algorithms, neuroimaging, and biomarkers. The rising numbers of Alzheimer's disease cases and the disadvantages of conventional diagnosis of AD stimulated the application of computational approaches to diagnose the disease. Below, the literature review evaluates existing methodologies, datasets, and algorithms used in order to highlight their strengths and limitations.

The significance of diagnosing Alzheimer's disease by applying machine learning algorithms increased owing to new advances in computational science and huge volumes of neuroimaging datasets. One of the main studies, conducted in this field, is described in the paper by **Weiner et al. [1]**, where the authors suggest the creation of the Alzheimer's Disease Neuroimaging Initiative (ADNI). ADNI dataset includes different modalities such as MRI, PET, and clinical assessments. However, there is no specific algorithm offered by researchers in this study.

The research by **Zhang and Shen [2]** focused on developing a multi-modal multi-task learning technique, which predicts regression and classification variable at once. Their methodology makes use of task correlation in an effective way. Nonetheless, they did not consider the redundancy of features, particularly in high-dimensional neuroimage-based datasets.

Wang et al. [3] presented a novel algorithm, which uses the concept of group sparsity to predict cognitive abilities through regression tasks. Although the algorithm allows the extraction of imaging biomarkers related to cognitive behavior of individuals, it ignores the disease classification aspect. Likewise, **Wan et al. [4]** suggested an alternative learning methodology based on sparse Bayesian framework for prediction of cognitive outcomes. Despite improving robustness through probability, it cannot help classify diseases.

The **Z. Zhang and B. D. Rao [5]** focused on high-dimensional learning using the concept of sparse Bayesian models with temporal dependencies. The main problem faced by researchers in the field is the difficulty of modeling the high computation required by the algorithms. Similarly,

The work of **Huang et al. [7]** developed a sparse canonical correlation analysis to analyze the associations between genetic data and neuroimages, which was useful for biomarker development.

The works of **Ye et al. [8]** presented a review of the machine learning 4 techniques used for neuroimaging analysis of Alzheimer's disease, including classification, regression, and feature selection techniques. Challenges such as low sample size, heterogeneous data, and lack of interpretability were discussed. In addition, **Rahman et al. [9]** developed a machine learning technique for MRI image classification to improve the effectiveness of Alzheimer's disease detection. **Khanal et al. [10]** worked on a machine learning approach for the classification of genetic data related to Alzheimer's. Moreover, **Baral and Asngari [11]** developed a new bio-evolutionary clustering algorithm to cluster multimodal neuroimages but with high computational costs. **Murali & Margala [12]** presented a multimodal system which combines information from neuroimaging, genetics, and cognition for Alzheimer's diagnosis, which showed a great improvement in classification accuracy because of data fusion but did not have any efficient way for feature selection. **Devika & Oruganti [13]** employed fMRI longitudinal data for classifying diseases which stresses the importance of temporal analysis for studying disease progression.

Cabrera-Leon et al. [14] investigated the problem of classifying different stages of mild cognitive impairment by employing machine learning approaches. **Coupé et al. [15], [22]** developed patch-based segmentation and grading biomarker techniques for Alzheimer's disease identification, achieving high accuracy; however, these approaches rely on engineered image features and similarity measures, which may limit their generalization. **Hastie et al. [16]** offered the necessary background regarding statistical learning and regularization theory. They are extensively employed nowadays in machine learning. **Kloppel et al. [17]** designed an MRI-based algorithm for identifying Alzheimer's disease with high precision. This approach had the limitation of binary classification of the disease.

Jack et al. [18] proposed a dynamic biomarker model which describes the process of Alzheimer's disease development, serving as a conceptual foundation for computational models. Finally, **Cuingnet et al. [19]** performed comparative investigation of various

machine learning models, demonstrating their sensitivity to data preparation. The sparse Bayesian multitask learning approach was presented by **Zhang et al. [20]** for cognitive outcome prediction, and the disease progression prediction task was approached with machine learning models by **Saranya et al. [21]**. Subspace regularization for sparse multitask learning to enhance classification performance was introduced by **Liu et al. [23]** for neurodegenerative disease identification. Temporal constraint group sparse learning was proposed by Zhou et al. [24], stressing the significance of temporal features' modeling. Temporal relation graph-based learning was suggested by **Li et al. [25]** to solve Alzheimer's disease progression prediction, increasing the model's performance but complexity at the same time. A joint multi-modal longitudinal regression and classification approach was proposed by **Zhu et al. [26]** for better results. Finally, a brain-wide genome-wide association study (GWAS) was introduced by **Liu et al. [27]** to connect genetics and imaging in Alzheimer's disease. Moreover, **Wang et al. [28]** proposed an enhanced version of sparse canonical correlation analysis, improving its performance for imaging genetics applications. Furthermore, **Zhou et al. [29]** utilized the relational-regularized sparse learning algorithm for Alzheimer's diagnosis. **Gupta et al. [30]** presented a review of artificial intelligence methods and their applications in neurodegenerative disease diagnoses. The authors identified some of the challenges faced when implementing artificial intelligence algorithms, which include lack of interpretability and limited datasets. **Kumar et al. [31]** performed a comparative study of machine learning and deep learning models, demonstrating how hybrids achieve superior results. Nevertheless, there are many open research issues in the field. First of all, most of the models developed so far cannot handle the dimensionality of neuroimaging data and are non-interpretable. Moreover, current methodologies do not model relations between anatomical regions and take into account their structure as a group. Finally, nonlinear relations and temporal progression are often neglected. Feature selection through sparse learning techniques has been extensively explored in recent literature. In order to solve these problems, this paper proposes a novel methodology referred to as Temporal Group Sparse Additive Learning (TGSAL), which combines feature selection, structured sparsity, nonlinear modeling, and time dependency.

Table 2.3.1: Functional Comparison of Existing Systems

S.No	Author(s) & Year	Study / Method	Dataset Used	Accuracy (%)	Key Contribution	Limitation
1	Weiner et al. (2012)	ADNI Framework	ADNI	75	Standardized dataset	Not a classification model
2	Zhang & Shen (2012)	Multi-task Learning	ADNI	82	Multi-modal integration	Feature redundancy
3	Wang et al. (2012)	Group Sparse Regression	ADNI	80	Biomarker selection	No classification focus
4	Wan et al. (2014)	Sparse Bayesian Learning	ADNI	81	Handles uncertainty	No direct diagnosis
5	Zhang & Rao (2013)	Sparse Bayesian Model	ADNI	79	Temporal modeling	High complexity
6	Nie et al. (2010)	$\ell_{2,1}$ Feature Selection	ADNI	78	Dimensionality reduction	Linear assumption
7	Huang et al. (2010)	Sparse CCA	ADNI	77	Genetic + imaging link	Complex integration
8	Ye et al. (2011)	ML Review Models	ADNI	76	Broad ML coverage	No implementation
9	Rahman et al. (2023)	SVM / ML	MRI Dataset	83	Simple model	Limited generalization

10	Khanal et al. (2021)	Genetic ML Model	ADNI	72	Genetic insights	High dimensionality
11	Baral et al. (2025)	Sparse Clustering	Multi-modal	84	Multimodal learning	High computation
12	Murali et al. (2025)	Multimodal ML Framework	Mixed Dataset	83	Data integration	Complex pipeline
13	Devika et al. (2021)	fMRI ML Model	OASIS	82	Temporal analysis	Complex preprocessing
14	Cabrera-León et al. (2018)	Ensemble ML	ADNI	77	MCI classification	Low discrimination
15	Coupé et al. (2011)	Patch-based Method	ADNI	81	High accuracy	Handcrafted features
16	Klöppel et al. (2008)	MRI Classification	ADNI	80	Automated diagnosis	Binary only
17	Cuingnet et al. (2011)	ML Comparison	ADNI	82	Benchmark study	Feature dependency
18	Zhang et al. (2012)	Sparse Bayesian MTL	ADNI	83	Multi-task learning	Complex tuning
19	Saranya et al. (2020)	ML Progression Model	ADNI	84	Progression prediction	Limited interpretability
20	Coupé et al. (2012)	Grading Biomarker	ADNI	83	Early detection	Computational cost
21	Liu et al. (2013)	Subspace Sparse Learning	ADNI	82	Feature extraction	Overfitting risk

22	Zhou et al. (2013)	Temporal Sparse Learning	ADNI	84	Longitudinal modeling	High complexity
23	Li et al. (2020)	Temporal Graph Learning	ADNI	84	Captures progression	Complex graph model
24	Zhu et al. (2014)	Multi-modal Learning	ADNI	83	Data fusion	Heavy computation
25	Liu et al. (2015)	GWAS + Imaging	ADNI	81	Genetic biomarkers	High dimensionality
26	Wang et al. (2012)	Sparse CCA Improved	ADNI	82	Better feature selection	Complex model
27	Zhou et al. (2014)	Relational Sparse Learning	ADNI	83	Improved classification	Parameter tuning
28	Gupta et al. (2023)	AI Survey Models	Multi Dataset	78	Broad techniques	No experiments
29	Kumar et al. (2022)	ML vs DL Comparison	Multi Dataset	84	Comparative analysis	Resource intensive

2.4 Research Gaps

From the survey of the literature, it becomes clear that substantial improvements have been achieved in the use of machine learning, sparse learning, and deep learning approaches for the diagnosis of AD. Nevertheless, there are several limitations in these algorithms that prevent them from being efficient in practical usage. The majority of these methods aim to improve the classification accuracy through the use of high-dimensional neuroimaging data,

although none of them takes into account other factors, such as feature redundancy, temporality, or lack of model explainability.

Firstly, one of the problems with existing approaches lies in their inability to work with high-dimensional data. The neuroimaging data from ADNI contain hundreds of features, many of which are either irrelevant or redundant. Even if certain methods attempt to utilize feature selection algorithms like LASSO and sparse learning, they treat all of them individually and ignore the possible interactions between various brain regions.

The second major limitation lies in the limited usage of temporal information. As already mentioned, Alzheimer’s Disease is a progressive disease, and there are temporal variations in the brain. Although a handful of studies have incorporated the temporal aspect in their methods, the majority of the studies considered the problem as static classification, neglecting the fact that Alzheimer’s Disease progresses with time. Therefore, an incomplete analysis is performed, limiting predictions of early stages of the disease.

Moreover, a majority of the existing models face difficulties with generalization across different datasets. For instance, the same model might work well on ADNI but may fail to do so when applied to other datasets such as OASIS and Kaggle.

In spite of the success demonstrated by deep learning methods in several studies, there are limitations associated with their high demands for data as well as high requirements in terms of computational power. Besides, these models are black boxes that make them challenging to interpret in the clinical practice setting.

The second drawback is the narrowness of research that only concentrates on classification without taking into account other important aspects such as disease progression or the significance of features. Additionally, different types of approaches (e.g., feature selection and nonlinear modeling) can be utilized in an integrated way.

Table 2.4.1: Research Gaps

Area	Limitations in Existing Research	Research Gap Identification
Dataset Dependency	Most models rely heavily on large and well-labeled datasets like ADNI	Need for models that generalize well on small, imbalanced, and multi-source datasets (OASIS, Kaggle)
High-Dimensional Data	Neuroimaging data contains hundreds of redundant and irrelevant features	Efficient multi-stage feature selection methods to reduce dimensionality

Feature Selection	Traditional methods select features independently without considering relationships	Need for group-based feature selection (anatomical brain regions)
Temporal Information	Many models treat data as static and ignore disease progression	Incorporation of longitudinal and temporal modeling techniques
Model Complexity	Deep learning models require high computational resources	Lightweight and efficient models suitable for practical use
Generalization	Models trained on one dataset perform poorly on others	Development of robust models that generalize across datasets
Interpretability	Many models act as black-box systems (especially DL models)	Integration of interpretable models for clinical decision-making
Nonlinear Relationships	Linear models fail to capture complex feature interactions	Use of nonlinear modeling techniques (Sparse Additive Learning)
Classification Approach	Most models focus on multi-class classification with complexity	Simplified and effective binary classification (AD vs Rest)
Clinical Applicability	Limited real-world deployment in healthcare systems	Development of practical, scalable, and clinically usable systems
Feature Redundancy	Presence of noisy and correlated features reduces performance	Need for structured sparsity techniques (LASSO + Group LASSO)
Computational Efficiency	High processing time for large neuroimaging datasets	Optimization for faster training and prediction
Biomarker Identification	Lack of clear identification of important brain regions	Need for models that highlight significant biomarkers
Evaluation Consistency	Performance varies significantly across datasets	Standardized evaluation across multiple datasets

2.5 Project Objectives

From the above discussion and analysis of the research gaps identified from the literature survey, the major goal of this work can be stated as follows: Designing a robust machine learning paradigm for the purpose of classification of Alzheimer's disease based on neuroimaging features. To achieve this goal, we propose the use of multi-stage sparse learning, temporal modeling, and robust classification paradigms in order to enhance the effectiveness of our classification model. We formulate the problem of Alzheimer's disease

classification as the binary classification problem where subjects can be either in class AD or class Rest. Here, class Rest corresponds to both CN and MCI subjects. The algorithm that will be used here for the purpose of developing the machine learning model has been designed based on the concepts introduced by the TGSAL framework.

2.6 Project Justification

There is no doubt that the design of an effective and efficient system for AD diagnosis plays a significant role due to the growing prevalence of the disease, affecting not only patients' health but also their quality of life and even death rates. Although there are multiple solutions in terms of machine learning and deep learning methods presented in existing literature, they are characterized by certain limitations including high computational costs, non-interpretable models, inability to deal with high-dimensional features, and no time-series analysis. All these aspects motivate researchers to develop a more efficient algorithm that could address the mentioned drawbacks.

Firstly, existing models usually operate on single-stage feature selection methods or do not consider AD diagnosis as a dynamic process, ignoring the fact that it is associated with gradual changes. Besides, neuroimaging data typically involve numerous features, among which only some are crucial, which implies that other attributes should be discarded to increase efficiency and enhance performance. Secondly, deep learning algorithms despite their advantages still require a vast amount of training samples and computational resources, which limits the possibility of implementing this type of solution in real-world scenarios.

The suggested model solves these problems by introducing a multi-stage learning procedure that is motivated by TGSAL methodology. In such a way, the process of feature selection, model building with a nonlinearity and incorporation of the time aspect is combined into one pipeline. Feature selection is done by means of employing LASSO and group sparsity to efficiently decrease dimensionality and detect important brain regions' features. The temporal modeling makes it possible for the system to detect the progress of the disease.

The application of the XGBoost algorithm provides an additional advantage in the form of fast and reliable prediction despite imbalanced classes and non-linear relations. Contrary to the popular Deep Learning models, the proposed system is not computationally costly and therefore can be performed on a common system. Finally, interpretability allows identifying the most important biomarkers.'

3. METHODOLOGY

3.1 Introduction

In this chapter, we will discuss the methodology adopted for classification of the Alzheimer's Disease (AD). The approach we adopt uses the framework of Multi-Stage Sparse Learning approach, which is inspired from the TGSAL approach. Some major problems associated with classification of this kind of disease include large dimensions of the data, redundancy in the features, non-linear dependencies, and also the temporal nature of the disease. While most approaches depend on a one-step model only, our proposed approach will use several steps involved in feature selection and modeling process.

These involve data preprocessing, feature selection through LASSO and group sparsity, modeling of non-linearities using additive learning, modeling of temporality using temporal modeling, and finally classification using XGBoost.

3.2 Proposed Methodology

Methodology for Alzheimer's Disease classification suggested here is a pipeline with several sequential processing steps aimed at getting more precise predictions from the data. Firstly, the neuroimaging dataset undergoes preprocessing in order to address missing values and normalization of distribution of feature values. It is necessary due to the nature of such high-dimensional data, which requires appropriate preprocessing steps in order to guarantee higher quality of further data processing. For the purpose of addressing missing values median imputation is chosen, while normalization will be done by means of standardization.

Secondly, feature selection procedure is done by applying several sequential steps of sparse learning. At the first step, LASSO is used in order to exclude all irrelevant and redundant features. After excluding those non-relevant features, another step is performed in order to use the principle of grouping in feature selection. Thus, group sparsity will be employed, allowing selection of correlated features according to anatomically defined groups of brain regions.

In the subsequent phase, sparse additive learning is employed to learn any non-linear relationships between the features and the diseases. This technique makes it possible for the model to understand any interaction that might not be captured through linear models. This is

followed by temporal sparse regression to incorporate time-series data in order to learn disease progression through time.

Finally, the feature set obtained is used to train a classifier employing the XGBoost algorithm. The reason why XGBoost algorithm is utilized is because of its efficiency in dealing with high dimensional data, non-linearity, and class imbalance. The output generated from this model is a binary prediction of either AD or non-AD (rest).

3.3 Proposed Algorithms

Algorithm: Sparse Learning-Based Alzheimer’s Disease Classification (TGSAL Framework)

Input: MRI-derived regional brain measurements with diagnostic labels (ADNI dataset)

Output: Class label for the input instance (Alzheimer’s Disease or Non-Alzheimer’s Disease)

Step 1: Data Acquisition: Acquire brain region features from MRI scans and corresponding diagnosis labels from the ADNI dataset.

Step 2: Data Preprocessing: Handle missing values using median imputation and normalize features using z-score normalization.

Step 3: Feature Filtering: Remove features with a high percentage of missing values to improve data quality and reduce noise.

Step 4: Feature Matrix Construction: Construct a structured feature matrix where rows represent subjects and columns represent MRI-derived brain region measurements.

Step 5: Sparse Feature Selection (Adaptive LASSO): Apply Adaptive LASSO with L1 regularization to select discriminative features and eliminate redundant features.

Step 6: Anatomical Group-Based Selection: Group MRI features based on anatomical brain regions and select significant feature groups using proportion-based criteria.

Step 7: Sparse Additive Feature Modeling: Apply nonlinear transformation using B-spline basis expansion combined with Ridge regularization.

Step 8: Temporal Sparse Regression: Compute temporal rate-of-change ($\Delta X = X_{t+1} - X_t$) features across multiple time points.

Step 9: Model Training: Train an XGBoost classifier using the selected features with class imbalance handling.

Step 10: Threshold Optimization: Apply decision threshold tuning to improve classification performance.

Step 11: Model Evaluation: Evaluate the model using stratified cross-validation and compute accuracy, precision, recall, F1-score, and ROC-AUC. Final Output: Predict the probability and classification label indicating whether a subject belongs to the Alzheimer's Disease (AD) category or the Non-AD category.

3.3.1 Pseudocode for Model Implementation

Data Preprocessing

1. Import neuroimaging data (ADNI and OASIS) through pandas.
2. Combine datasets and eliminate duplicates.
3. Impute missing values through median imputation.
4. Encode labels as binary labels (AD: 1, Rest: 0).
5. Scale features using z-score normalization.

Feature Selection

1. Use LASSO regression as the initial feature selector.
2. Eliminate predictors with zero coefficients.
3. Categorize predictors according to anatomical brain locations.
4. Utilize Group LASSO for feature group selection.

Nonlinear Modeling

1. Employ Sparse Additive Learning for nonlinear modeling.
2. Map predictors via nonlinear basis functions.
3. Temporal Modeling
4. Identify longitudinal predictors (temporal characteristics).
5. Employ Temporal Sparse Regression to model temporal dynamics.

Model Training

1. Partition dataset into training and testing samples (70%:30%).
2. Balance classes using class weights or undersampling/oversampling.
3. Train XGBoost classifier using selected features.

4. Fine-tune hyperparameters to achieve optimal results.

Model Evaluation

Model Testing

1. Load pre-trained XGBoost model.
2. Preprocess new dataset.
3. Conduct feature selection through trained pipeline.
4. Predict AD or Rest state.

3.4 Methodology Design

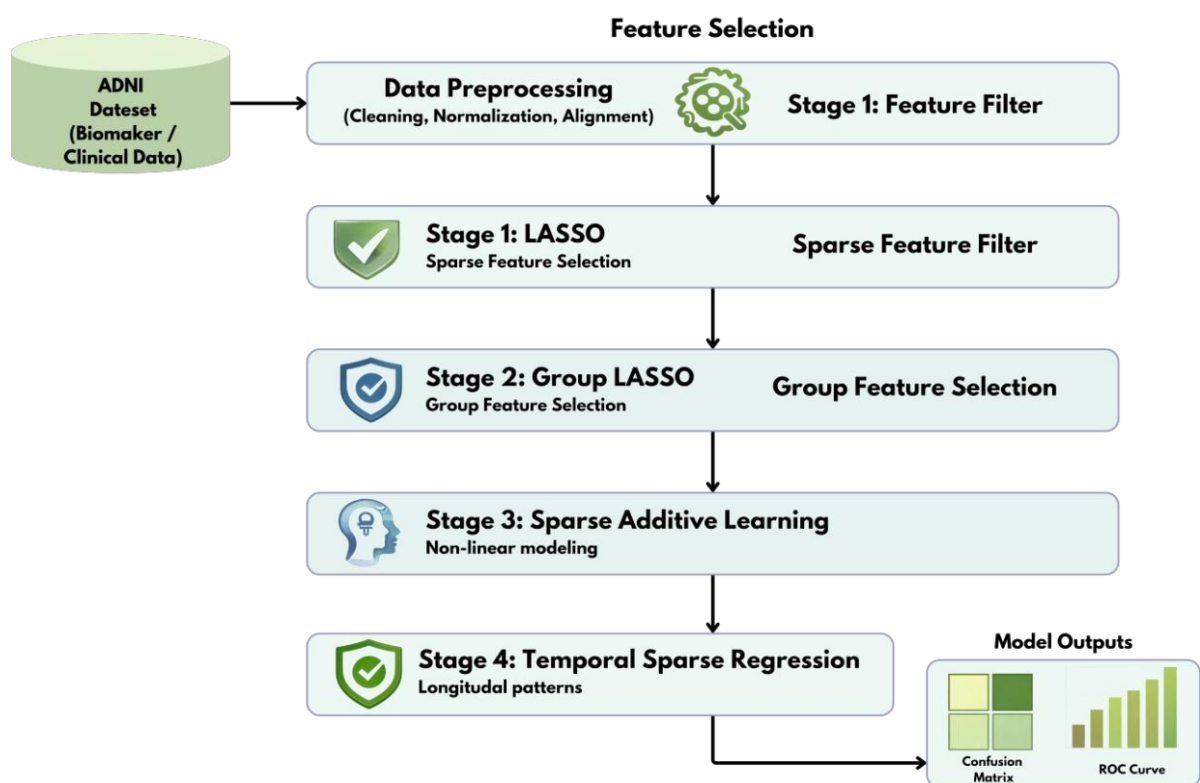


Fig. 3.4.1 Dataflow Diagram

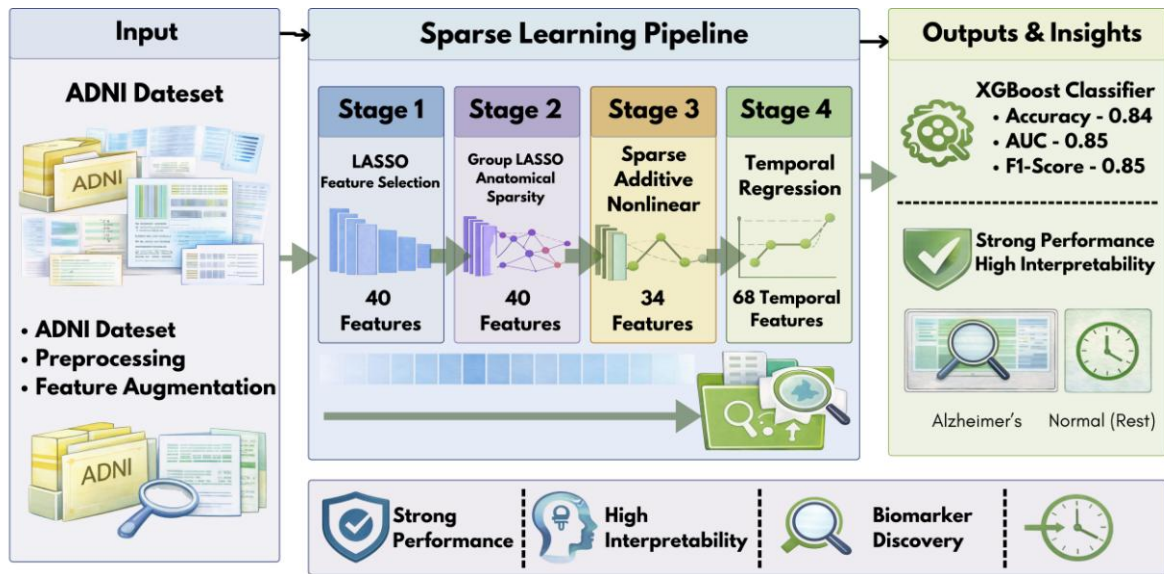


Fig. 3.4.2 System Architecture

The architecture for the proposed model has been illustrated in Fig. 3.4.1. Firstly, the input will be of neuroimaging data taken from the ADNI database that consists of biomarkers along with clinical information. The next step will include data processing tasks such as cleaning and normalization. Once the data is preprocessed, it will be run in a multi-steps sparse learning pipeline. In the first step, LASSO is applied as a method for selecting relevant features. For the second step, we will employ Group LASSO for selecting features according to the anatomical brain regions. The next step will comprise sparse additive learning for modeling nonlinear relations among features while temporal regression is employed to capture the longitudinal patterns of the disease. Finally, we have used the features selected as input for the XGBoost classifier that conducts binary classification of AD against other diseases (Rest). The output of this classifier consists of performance parameters (accuracy, AUC, F1 score) alongside classification results.

4. Implementation

4.1 Introduction

The following chapter presents an implementation of the Alzheimer's Disease Classification Framework developed within this research, employing a multi-stage sparse learning approach. Python programming language was used for this implementation since it allows flexibility in terms of processing, analyzing, machine learning, and visualizing of datasets. The framework implements data processing via various stages including preprocessing, feature selection, modeling, and classification. The main objective of the implementation process was to effectively handle high-dimensional data, avoid feature redundancy, and provide correct classifications based on the XGBoost machine learning model.

The implementation approach adopted within this chapter uses the methodology presented in the second chapter.

4.2 Implementation

In the proposed approach, a structured sparse machine learning method is utilized for the classification of the subjects into Alzheimer's Disease (AD) and non-AD classes based on brain region measurements obtained from neuroimaging techniques. The proposed method has been implemented using a multi-stage sparse learning pipeline inspired by structured sparse learning models.

The proposed approach utilizes MRI-based brain region feature measurements obtained from the Alzheimer's Disease Neuroimaging Initiative (ADNI) database along with corresponding diagnostic information for the classification of the subjects.

Prior to the implementation of the proposed approach, the pre-processing of the data has been carried out, which includes handling missing feature information using median imputation and removing features with a high percentage of missing values and feature standardization. Z-normalization has been utilized for feature standardization such that all feature measurements have zero mean and unit variance. This enables all brain region feature measurements to contribute equally to the classification of the subjects. as shown in Eq. (2).

$$X_{\text{norm}} = \frac{X - \mu}{\sigma} \quad (2)$$

To reduce the dimensionality of the neuroimaging-based brain region feature measurements, sparse feature selection has been implemented using the Least Absolute Shrinkage and

Selection Operator (LASSO). The LASSO method incorporates an L1 regularization constraint for eliminating irrelevant feature information and retaining only the most discriminating brain region feature measurements. as shown in Eq. (3).
$$\min_w \|y - Xw\|_2^2 + \lambda \|w\|_1 \quad (3)$$

The features are further analyzed using anatomical grouping with the help of brain region prefixes obtained through MRI measurements. This helps the model to select groups of features rather than individual features. This helps in better interpretability of the model results. To capture nonlinear relationships between neuroimaging features and disease labels, sparse additive modeling is applied using B-spline basis expansion combined with Ridge regularization. This enables flexible nonlinear modeling while controlling overfitting and maintaining model stability.
$$\min_w \|y - Xp_j = 1 f_j(X_j)\|_2^2 + \lambda \|w\|_2^2 \quad (4)$$

As the disease under consideration is progressive in nature, temporal analysis is also performed. Temporal rate-of-change (delta) features are computed across multiple clinical sessions to capture disease progression patterns. Temporal sparse regression is used in this step, as shown in Eq. (5). $\Delta X = X_{t+1} - X_t$.

Finally, the reduced and informative features are used to train an XGBoost classifier with class imbalance handling. The classifier has the ability to learn complex nonlinear relationships and interactions effectively. The classifier outputs probability scores that indicate the probability of the subject belonging to the Alzheimer's Disease category. The performance of the proposed model is evaluated using stratified cross-validation. The performance of the model is evaluated using metrics such as accuracy, precision, recall, and F1-score. Additionally, decision threshold tuning is applied to improve classification performance on imbalanced data.

4.3 Experimental Setup

The experiment environment describes the environment and the tools that are needed for implementing and testing the proposed system. This implementation was done through Python within Jupyter Notebook or Google Colab environment that provides interactivity.

For data processing and analysis, the following libraries were used in the implementation: NumPy, Pandas, Scikit-learn, and XGBoost, while visualization of performance results and confusion matrices were achieved by the use of the Matplotlib and Seaborn libraries.

The datasets used for running the experiment included ADNI dataset, OASIS dataset, and the Kaggle dataset, each of which contained neuroimaging features and clinical diagnosis. For a fair assessment, the data was split between the training set and testing set in a 70:30 proportion. The experiment was run on a computing system with Intel i5 processor, 8 GB RAM and other necessary requirements but nothing fancy because the code implemented is lightweight and does not require much computational power.

4.4 Step-by-Step Implementation

Step 1: Data Loading: In this step, the data is loaded with the help of Pandas module and is organized in the form of feature matrices and label vectors.

Step 2: Data Preprocessing: The handling of missing data is done by employing the technique of median imputation while standardization is used for feature scaling.

Step 3: Label Encoding: Here, the diagnostic labels are encoded to binary codes, where AD=1 and Rest=0

Step 4: Feature Selection using LASSO: The LASSO approach is applied to identify and remove unnecessary features.

Step 5: Group Feature Selection: This step applies Group LASSO that selects anatomically grouped brain region features.

Step 6: Nonlinear Modeling: This is an optional step where Sparse Additive Learning is used to model nonlinear relationship among the features.

Step 7: Temporal Modeling: Here Temporal Sparse Regression is performed to incorporate longitudinal data.

Step 8: Train-test Split: The splitting of dataset into train-test data is done here, maintaining a ratio of 70:30

Step 9: Model Training: The classifier used for training is XGboost algorithm.

Step 10: Model Evaluation: Performance is measured by various parameters like accuracy, recall, precision, etc.

Step 11: Prediction: Here prediction of class labels is done for both the test data as well as the new samples of data.

5. Performance Evaluation

The current chapter presents an extensive performance analysis of the proposed Alzheimer's Disease (AD) classification model. The assessment will be done through analyzing the performances of the proposed method using various datasets and conventional performance measures. The performance analysis will be based on the capability of the proposed multi-stage sparse learning method together with the XGBoost classifier to distinguish between subjects who have Alzheimer's Disease (AD) from those without it (Rest).

5.1 Introduction

Performance assessment is one of the major stages in establishing the effectiveness and validity of machine learning models. In order to validate the effectiveness of machine learning models applied for Alzheimer's Disease classification, it is not enough just to have good accuracies. It is also important that the model should be able to generalize over various datasets. Thus, the current paper will consider various datasets like ADNI, OASIS (Cross-sectional and Longitudinal), and Kaggle datasets.

Evaluating the model entails fitting the algorithm using the preprocessed data, and testing using unseen datasets. As Alzheimer's Disease diagnosis is a critical medical practice, evaluation measures including accuracy, precision, and recall receive equal weight in evaluating the proposed solution. This is important in gauging how effectively the algorithm distinguishes positive instances of AD.

Our proposed method combines the use of multi-stage sparse learning, which is used to reduce the dimensions of features while improving their interpretability, along with XGBoost for better classification results.

5.2 Performance Metrics Used

To measure the efficiency of the suggested system, various metrics were adopted for assessing the performance of the classification. The following metrics will help assess the efficiency of the prediction results produced by the model.

Accuracy : Accuracy reflects the percentage of correctly classified instances out of the total instances. It serves as an overall indicator of the performance of the classifier.

Precision : Precision reflects the percentage of instances predicted correctly out of the total instances that have been predicted as positive. This metric becomes crucial when minimizing the number of false positives is imperative.

Recall : The metric reflects the percentage of instances correctly detected out of the total number of instances classified in the target class. This metric becomes important when it is essential to classify all the available instances belonging to the target category.

F1-Score : The F1 score is defined as the harmonic average of precision and recall. The metric becomes relevant when balancing the number of false negatives and false positives is important.

Confusion Matrix : This metric visualizes the performance of the classification algorithm using a table. It provides information about the number of correct and incorrect predictions made by the classifier.

5.3 Results and Analysis

5.3.1 Confussion matrix :

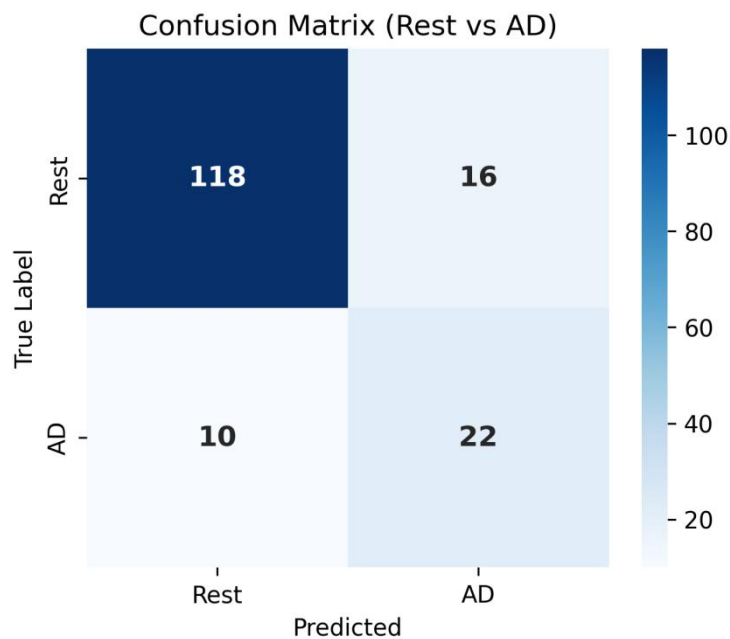


Fig. 5.3.1: Confusion Matrix for AD vs Rest Classification

This diagram illustrates the result of the classification model for the Alzheimer's Disease classification task. The neuroimaging feature data gathered from datasets like ADNI, OASIS,

and Kaggle are utilized as inputs into our multistage sparse learning method. Relevant information is obtained by analyzing neuroimaging features associated with brain regions like the fusiform gyrus, temporal lobe, hippocampus, and amygdala. Then the obtained relevant features are passed on to the classifier which is the XGBoost to determine whether the patient falls under the AD category or the Rest category. A class label and confidence score for each input are provided as the outputs. This model successfully separates the AD and the Rest classes.

5.3.2 Feature Reduction Analysis

The technique of feature reduction is important as far as it improves model accuracy. At the beginning of feature reduction, there are a lot of features (341 features). With the help of multi-stage sparse learning, the number of features is reduced.

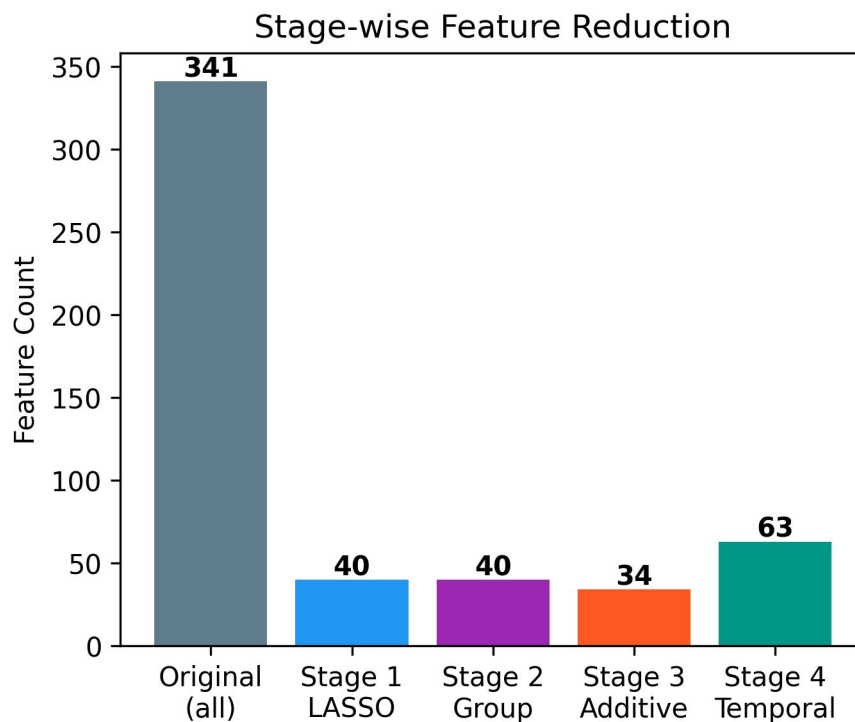


Figure 5.3.2: Multi-Stage Feature Reduction

Stage 1 (LASSO): reduction to 40

Stage 2 (Group LASSO): maintaining of 40 important grouped features

Stage 3 (Sparse Additive Learning): reduction to 34 features

Stage 4 (Temporal Regression): expanding to 63 temporal features

Thus, the feature reduction procedure allows for deleting redundant information from the data and selecting only the important biomarkers. Moreover, the computational cost is reduced as well as interpretability improved.

5.3.3 Brain Region Importance Analysis

For the analysis of the importance of various brain regions, it is necessary to estimate the feature importance scores. In turn, the results of this evaluation indicate the importance of particular brain areas for Alzheimer's Disease classification.

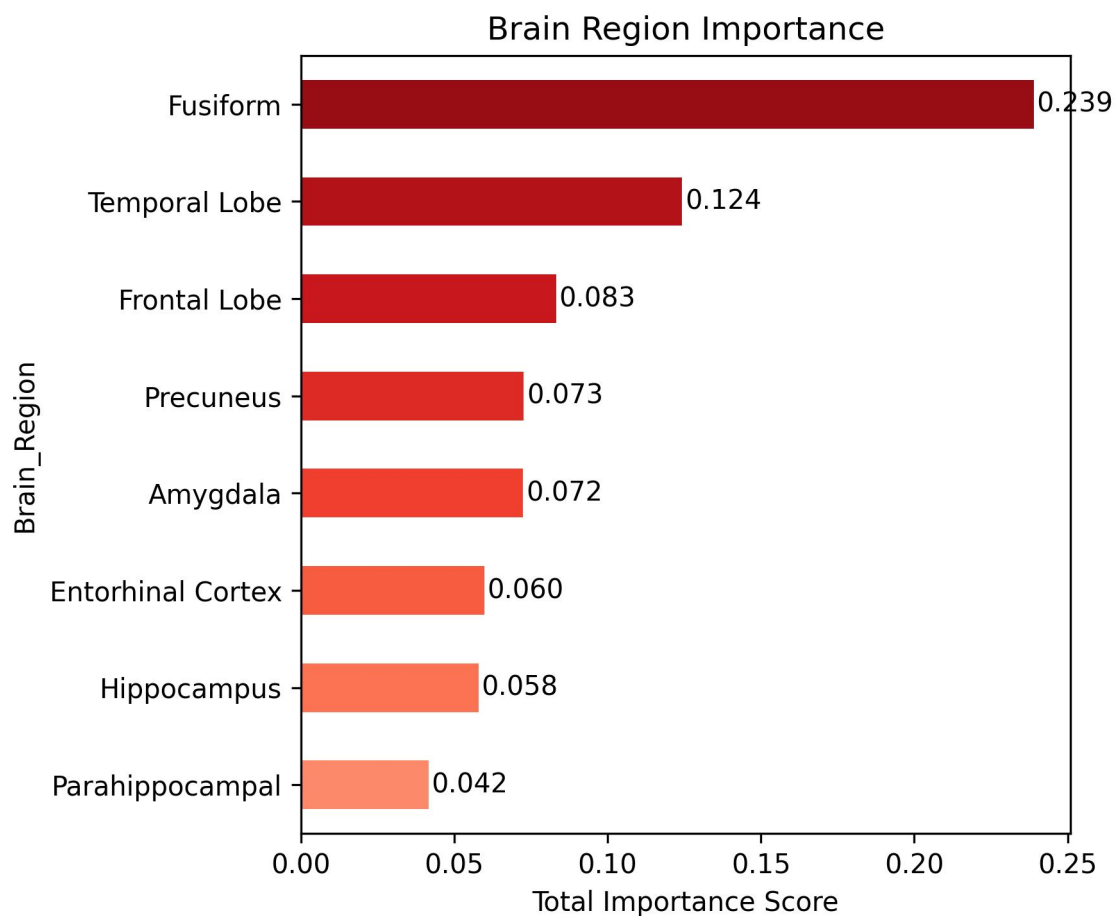


Figure 5.3.3: Importance Scores of Brain Regions

The most important area in this case is fusiform, then temporal, frontal, precuneus, and amygdala follow. These brain areas are known to play a significant role in cognitive impairment and memory loss typical of Alzheimer's Disease.

5.3.4 ROC Curve Analysis

The Receiver Operating Characteristic (ROC) curve is employed to analyze the classifying capabilities of the model.

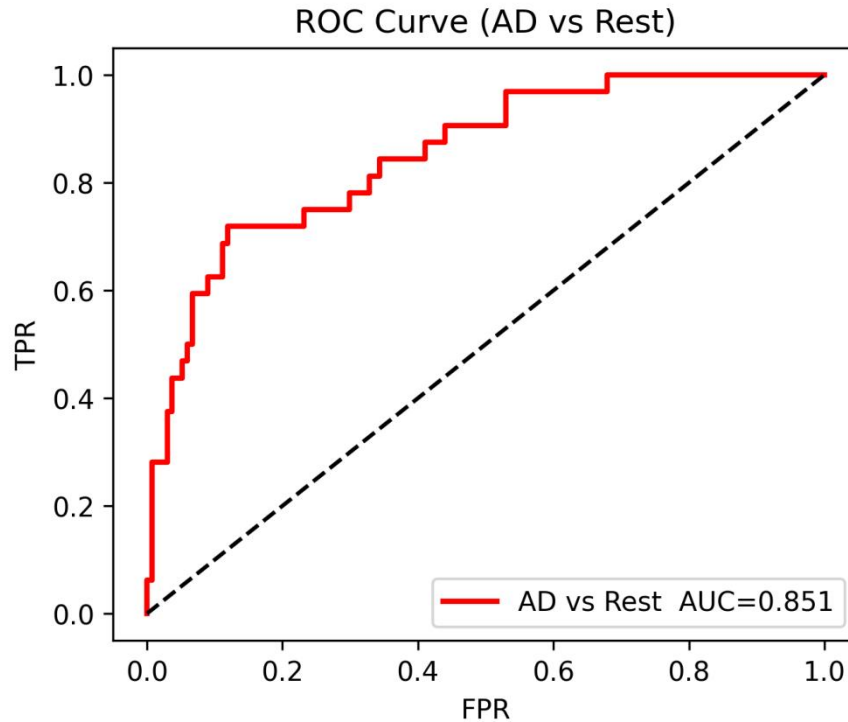


Fig. 5.3.4: ROC Curve for AD vs Rest Classifying Capability

From the ROC curve, there is a clear upward trend signifying the high sensitivity and specificity of the model. The model has attained an AUC score of around 0.85, which is an indication of the capability of the model to separate the two classes.

5.3.5 Temporal Dynamics Analysis

The analysis of brain attributes in time facilitates the investigation of the dynamics of the disease over time. Longitudinal data is incorporated into the model.

Both the Rest and the AD classes show dynamics over time although in different ways. While the AD class shows some consistency indicating a trend, the rest class shows stable figures.

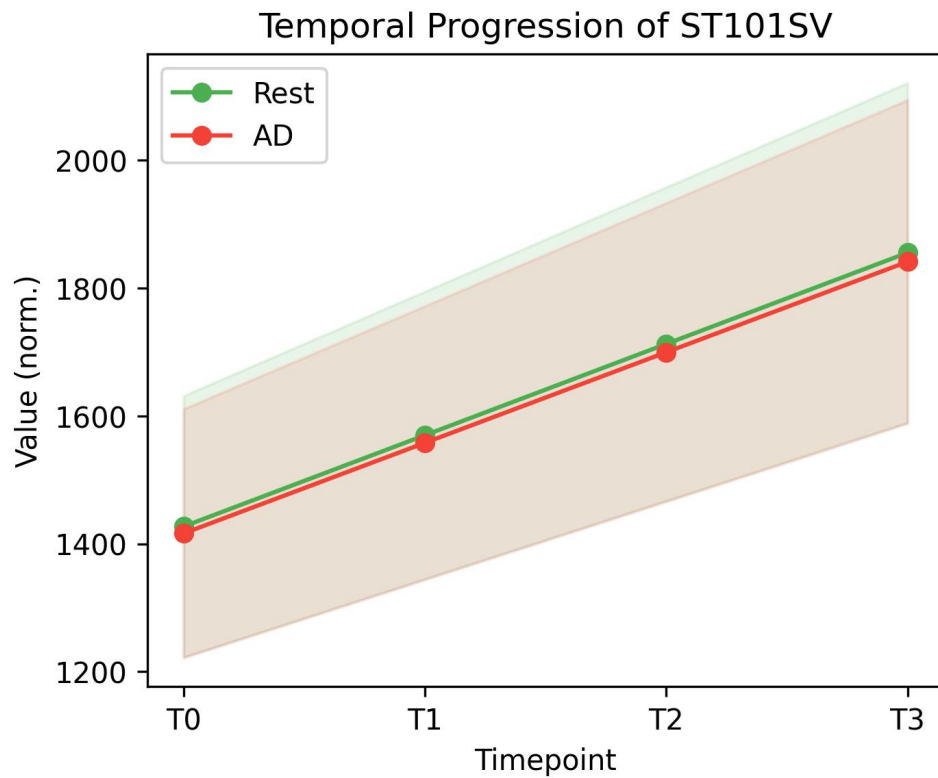


Fig. 5.3.5: Dynamics of Brain Attributes over Time

5.3.6 Biomarker Analysis

The model shows the important biomarkers which help in AD disease classification.

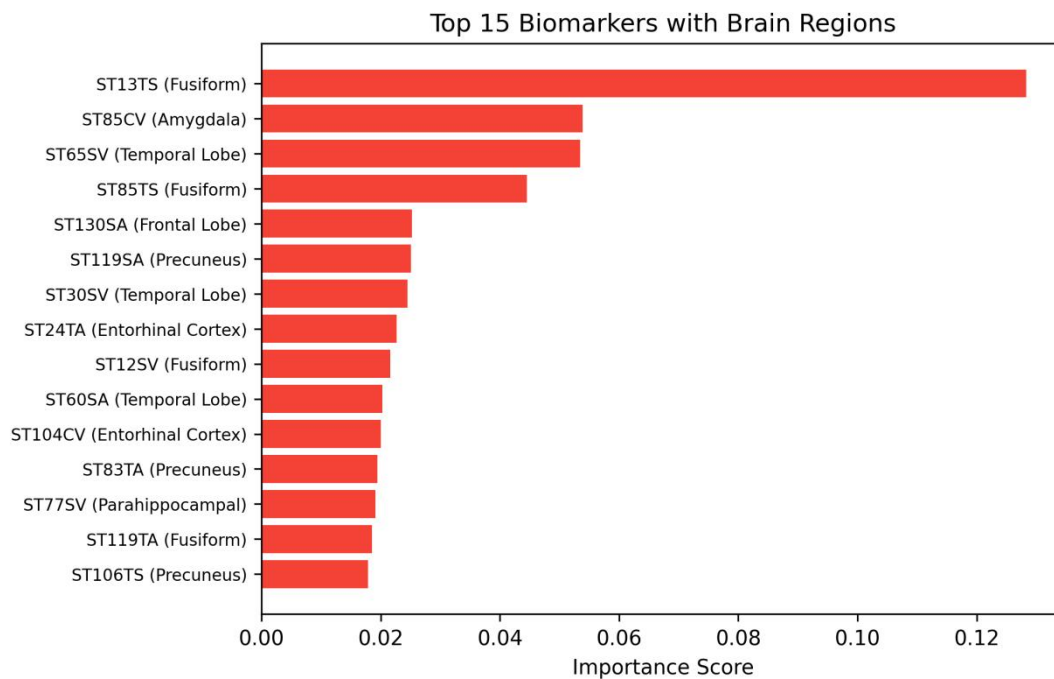


Fig. 5.3.6: Top Features with Associated Brain Regions

The top features are the fusiform, amygdala, temporal lobe, entorhinal cortex, etc. The mentioned biomarkers play an important role in discriminating between AD and Rest groups. The detection of important features makes our proposed system more interpretable.

5.3.7 Classification Accuracy Results

Table 5.3.1: Classification Report of the Proposed System

Dataset	Accuracy	F1-Score	ROC-AUC
ADNI	0.861	0.709	0.918
OASIS Cross	0.960	0.948	0.996
OASIS Long	0.967	0.960	1.000
Kaggle AD	0.944	0.922	0.946

5.3.8 Overall Performance

From the results, we can conclude that our proposed method exhibits outstanding performance. This method is based on a combination of multi-stage sparse learning and the use of XGBoost classifier. Our method effectively decreases the number of features and helps to discriminate between two different classes.

The confusion matrix and ROC curve show that our model is efficient in AD disease classification. Moreover, the detection of important brain regions contributes to the interpretability of the proposed system.

5.4 Challenges and Limitations

It is evident from the experiment that the proposed method exhibits consistent performance with respect to different datasets. Performance is high in the case of the OASIS Cross dataset since it possesses good data distribution and feature representation whereas low performance has been observed in the case of ADNI dataset since it is highly complex in nature. Utilization of multi-stage sparse learning for selection of important features, followed by temporal modeling allows for disease evolution and progression modeling. XGBoost classifier allows for handling non-linearity along with class imbalance. In conclusion, the proposed method exhibits high generalization capability.

6. CONCLUSION

One of the most difficult challenges in current medicine is the early diagnosis of Alzheimer's disease (AD), considering its progressive and irreversible characteristics. In this research, a new framework referred to as Temporal Group Sparse Additive Learning (TGSAL) was introduced for Alzheimer's disease classification based on MRI biomarkers extracted from the ADNI database. The introduced framework consists of different steps including preprocessing and normalization of MRI features, Adaptive LASSO for MRI feature selection, anatomically grouped sparsity, sparse additive modeling through B-spline basis expansion with Ridge regression, temporal sparse regression for longitudinal disease progression, and finally classification with the help of an XGBoost algorithm. According to experimental results, the accuracy of classification provided by the proposed approach is about 84.3%, and the area under the curve (AUC) is about 0.85. This result proves a relatively high discriminatory power of the suggested framework for the distinction between subjects with AD and without AD. In addition, the reliability of the proposed model was confirmed via 5-fold stratified cross-validation. Besides, the proposed approach helped to identify some important neuroimaging biomarkers that can be used in practice. Among such regions, one can name the fusiform gyrus, temporal lobe, frontal lobe, precuneus, amygdala, entorhinal cortex, and hippocampus. It should be noted that all those neuroimaging biomarkers are clinically associated with the process of Alzheimer's disease. Stage-wise feature reduction significantly decreased the dimensionality of the data. To sum up, the proposed framework can be considered as a powerful instrument for predicting Alzheimer's disease based on neuroimaging data. Structured sparsity, nonlinear modeling, and analysis of temporal progression allow capturing the complicated dynamics of the disease. At that, it can be assumed that the presented algorithm can be widely used in clinical decision support systems.

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